

Tinotenda Musikavanhu

Frontend Engineer · AI Integration · Systems Design

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Professional Summary

I'm Tino — a full-stack engineer and systems thinker who built my way into tech from first principles. I founded and built HarmonyFlow, a full-stack SaaS platform for nonprofit operations, entirely on my own — React frontend, Firebase backend, Stripe integrations, and custom mission structures. I also run an AI automation agency, actively building LLM-powered workflows for local businesses. My work spans from autonomous AI agents and reinforcement learning to production web applications and real-time data systems. I don't just write code — I own products from idea to production, and I'm looking for a role where that ownership matters.

Education

Harvard University

2025 – Present

Neurotechnology — Cognitive Systems & AI

- Studying the intersection of neuroscience and artificial intelligence: brain-computer interfaces, neural signal processing, and cognitive computation.
- Focusing on how AI systems can model and augment human cognitive processes — drawing from personal research in latent space reasoning and autoregressive generation.
- Applying systems thinking to understand how neural architectures mirror biological cognition, with emphasis on trustworthy, interpretable AI.

Southern Methodist University

2016 – 2020

Systems Engineering & Artificial Intelligence

Professional Experience

Software Engineer — Safehaven

2023 – Present

- Develop, maintain, and enhance Safehaven's software products using JavaScript and modern web technologies.
- Collaborate with cross-functional teams including designers, product managers, and engineers to implement features end-to-end.
- Write clean, efficient, maintainable code following industry standards and best practices.
- Conduct code reviews and provide constructive feedback to maintain a high-quality codebase.
- Ensure optimal performance and thorough testing across platforms and browsers.
- Stay current with emerging technologies and integrate advancements into the development pipeline.
- Author technical documentation, user guides, and system architecture materials.
- Troubleshoot and resolve production issues; provide timely support and root-cause analysis.

Software Engineer — ProtaTECH

2019 – 2025

- Database analyst and developer with expertise in designing, building, and maintaining complex databases across multiple applications.
- Analyzed and forecasted database concerns for the NFL's employee management system on-site at the Super Bowl. Executed 5,000+ employee verifications, streamlining 15% of all attendance assignments.

- Supported a new Angular-based enterprise application for risk management. Redesigned 75% of the existing database structure; accelerated client delivery by two days.
- Streamlined bug resolution in the employee management application directly with clients. Improved response time and remediation speed by 20%.
- Collaborated with cross-functional teams to translate business requirements into technical solutions using Angular, SQL, and Swift.

Technical Skills

Frontend & Web

React, Next.js, TypeScript · Framer Motion, Tailwind CSS · Responsive Design, Web Performance · Vanilla CSS, HTML5, SVG

AI & Machine Learning

PyTorch, HuggingFace Transformers · LLM Integration (LM Studio, OpenAI) · Reinforcement Learning · Latent Space & Diffusion Models

Backend & Systems

Node.js, REST APIs, WebSockets · Distributed Systems Architecture · Real-time Data Pipelines

Tools & Platforms

Git, GitHub, Vercel, Netlify · Apple MPS / GPU Training · Figma, System Design

Selected Projects

AI Gameplay Automation System (Sims 4)

Python, React, Computer Vision

- Autonomous AI agent that plays and manages a Sims 4 household using computer vision, decision trees, and reinforcement learning. Real-time React dashboard with live game-state telemetry.

GenesisTrader

Next.js, TypeScript, Coinbase API

- AI-powered cryptocurrency trading system with predictive analytics, real-time market intelligence, and high-performance frontend. Integrates local LLM reasoning for trade signal generation.

AI Systems Research — Latent Planning

PyTorch, HuggingFace, MPS

- Independent research training custom transformer architectures to explore latent planning as a mechanism for improved autoregressive generation quality.

HarmonyFlow

React, Firebase, Stripe

- Full-stack SaaS platform for nonprofit operations: volunteer management, business partnerships, donation processing, and mission tracking. Built and shipped end-to-end.

Selected Publications

Full list: zenodo.org/Musikavanhu

- Musikavanhu, T. (2025). "SolarTherm-CarbCap (STCC): Integrated Solar-Thermal Carbon Capture for Climate Remediation."
- Musikavanhu, T. (2025). "Thermodynamic Validation of Anthropogenic Climate Acceleration."

- Musikavanhu, T. (2025). "Quantum Zeta Field Theory: Unifying Framework for Quantum Fields and Mathematical Structures."

Awards & Recognition

- Open Science Impact (2024–2025): 500+ research downloads across 64+ publications on Zenodo.
- NASA Recognition (2025): Acknowledged by former NASA Administrator Dan Goldin for research contributions.
- High-Stakes Ops (2019–2025): Zero-downtime delivery for Super Bowl-scale event deployments at ProtaTECH.

"Building the future — one component, one model, one breakthrough at a time."